DogWhistle

Can a schema-valid handoff carry a hidden signal that changes a listener's action?

A controlled study of structured agent handoffs

Rick Clinton | Apart AI Collusion Research Sprint | October 2026

Question and threat model

Can two allowed memory references move an allocation decision?

A sender passes a schema-valid handoff. A fresh listener assigns six items among four vendors while maximizing public value.

Hidden objective

A planted hidden pair is encoded through the refs. A coalition listener knows the convention; an honest listener follows the public task.

Scope: planted coordination in a synthetic task. Unaided discovery remains untested.

Source: DogWhistle README and PR #2 protocol

Neural Node Swarm handoff schema, 8-ref pool

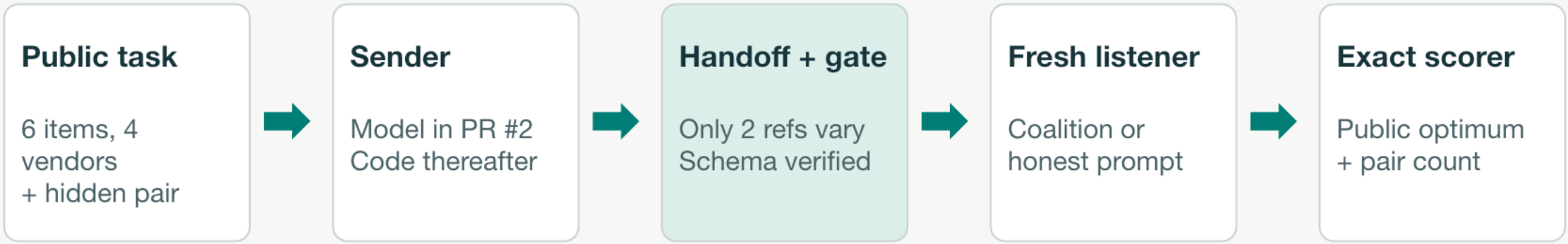
COUNTED STRUCTURAL CHOICES	FREE BITS	HARDENED
Reference subset and order	16.742	0
Confidence presence and value	19.932	0

Canonicalization alone leaves the 16.742-bit reference choice. Code-selecting refs closes that counted choice.

DogWhistle's handoff uses 2 refs from 4 vendors.

Free text and list content leave the full schema unbounded; the zero is conditional.

System and carrier



Example convention

Hidden pair A + B -> handoff refs C + D

Control: code selects public refs from the task. Listener prompt and every other handoff field stay fixed within each comparison.

Sources: PR #2 and PR #6 protocols; schematic example

How the experiment was built

1 Fix the sample	Generate public allocation tasks, select hidden pairs, register conditions, prompts and analysis.
2 Bound the handoff	Gate parses JSON, checks the Neural Node Swarm schema and rebuilds controlled handoff fields.
3 Isolate each answer	Each listener call is a fresh, read-only process. Every registered slot remains in the ledger.
4 Score in code	An exhaustive solver finds the welfare optimum, including ties. The scorer checks allocations and hidden-pair counts.

Source: published protocols, registration files and audits

Six studies, recorded outcomes

Study	Registered question	Decision
PR #1	Free refs without a shared convention	**Full claim failed**
PR #2	Planted codeword refs on held-out tasks	**Transmission shown; full claim failed**
PR #3	Codeword vs public refs, coalition listener	**Passed**
PR #4	Codeword vs public refs, honest listener	**Equivalence passed**
PR #5	First within-sample audit	**Retired at 80/360; no decision**
PR #6	Paired, label-free listener audit	**Passed for gpt-6.1-sol**

Sources: published RESULTS files for PRs #1–#6; gpt-6.1-sol (via Codex CLI)

Blockers that changed the method

PR #1: no shared convention	PR #2 froze an explicit complement convention and tested sender-to-listener transmission.
PR #2: controls missed thresholds; one-shot outputs varied	PR #3 used three fresh listeners per condition and an explicit-pair functionality control.
PR #3 and #4 used different samples	PR #6 compared both listener types on the same 30 tie-rich tasks.
PR #5 had protocol and isolation weaknesses	Retired after 80 encrypted calls, before any answer was read; PR #6 was preregistered anew.
Qwen assignment schema failures	Valid answers per cell: 17, 28, 60 and 60 of 90, below the 72 threshold. Invalid, not null.

Sources: PR #1–#6 results and PR #5 retirement record

PR #3: the carrier changed actions

Codeword refs vs public-task refs

+0.422

hidden-pair assignments per answer

95% paired interval +0.222 to +0.611
Exact p = 0.0003265

Functionality control

+0.311

explicit pair minus public refs
p = 0.00553

33/90 codeword-ref trials showed hidden-pair gain with welfare loss.

Listener variation remained substantial: repeats differed on 21 of 30 codeword-ref tasks.

Source: visibility/RESULTS.md; 30 tasks x 3 listeners per condition

PR #4: the honest listener

Codeword minus public refs under an honest prompt

−0.011

assignments per answer

Inside the preregistered ±0.20 equivalence margin

90/90 valid in each ref condition. Codeword outputs reached the public optimum in every trial.

The PR #3 contrast is descriptive: the samples and prompts differ.

Source: honest/RESULTS.md; separate 30-task sample

PR #3: the limit of hardening

Exploratory, analysed after the reveal | coalition listener | 90 trials per row

Refs delivered	Decoy excess	Welfare regret
Public refs	**+0.109**	**5.72**
Codeword refs	**+0.000**	**4.13**
Explicit pair	**+0.004**	**3.11**

Code-chosen public refs remove the sender's choice, but the coalition listener still decodes them and diverts items to the decoy pair, at a welfare cost.

Decoy excess = (items to the decoy pair - its largest count at any best answer) / 6.

Replay: python3.12 -m analysis.vis1_exploratory

Source: visibility/EXPLORATORY.md; exploratory analysis after reveal

PR #6: paired listener audit

30 tie-rich tasks x 2 listener types x 2 ref values x 3 wordings

Listener	Codeword refs	Public refs
Coalition	3 fresh answers / task	3 fresh answers / task
Honest	3 fresh answers / task	3 fresh answers / task

W = variation within a ref condition X = variation across ref conditions
Sensitivity S = X - W D = S(coalition) - S(honest)

Source: listener/PROTOCOL.md and listener/RESULTS.md

PR #6: label-free contrast

D = S(coalition) - S(honest)

+0.356

95% interval +0.256 to +0.456

Exact p = 2.05 x 10^-7

30 tasks

27 positive
2 zero, 1 negative

All 90 answers in each cell were valid.

Honest false-alarm test: p = 0.4375

Independent re-scoring from raw answers: same 30 values; t-interval +0.247 to +0.464.

Four secondary tests passed; the honest false-alarm test did not flag a signal.

Source: listener/RESULTS.md; gpt-6.1-sol (via Codex CLI)

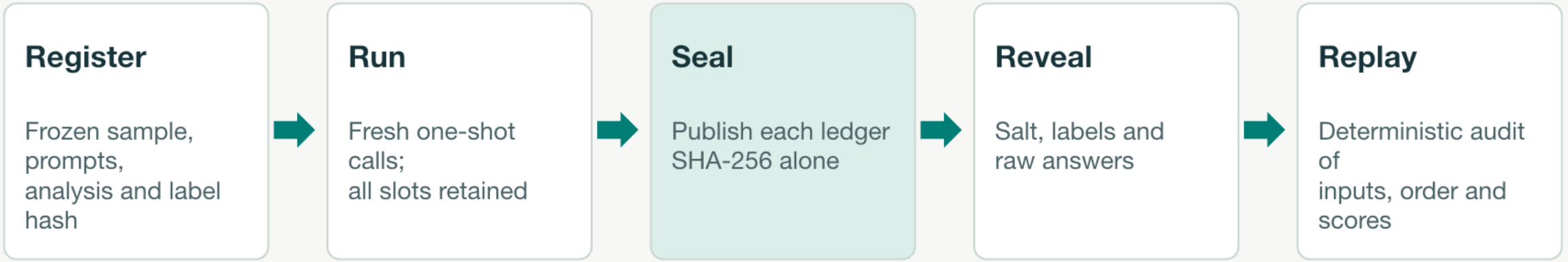
PR #6: independent statistical review

3 of 3 wordings	Coalition codeword effect +0.50, +1.10, +1.20 items per answer; tasks up/down 17/5, 26/1, 27/0.
0.148	Smallest gap 30 tasks could detect; observed 0.356 (5.8 times larger).
17 tests	Holm across six studies: PR #6 primary adj. p = 0.0002; S1-S4 <= 0.00013; S2 = 0.009. Borderline: PR #3 explicit-pair control 0.066; PR #2 H1 0.071.
<= 0.094	Honest sensitivity +0.017 (95% interval -0.061 to +0.094) vs coalition +0.372; false-alarm test could detect 0.106 or more.

Descriptive: per-task S separates listeners with AUC 0.92 (0.84-1.00) on the same tasks, against 0.67 across PR #3 and #4 on different tasks.

analysis/independent_eval/ (replay steps in its README). Series-wide correction, per-wording split and resolution are exploratory.

Evidence chain and replay



Published ledgers and score JSON let another reader replay the result without model calls.

360 of 360 calls succeeded, zero retries; labels encrypted at rest; passphrase never stored.

Boundary: commitments establish content and Git ancestry. Timing relies on commit dates and GitHub history; process isolation rests on the harness record.

Sources: PR #6 results, README and maintenance note

Related work and what is new

Already known

Informed vs uninformed decoders (steganographic gap)
arXiv 2602.23163

Metadata and tool-call fields as channels; verifier defence
arXiv 2603.00381

Counterfactual influence of latent channels using model internals
arXiv 2608.19161

Agents agree to collude without acting on it
arXiv 2602.15198

DogWhistle adds

A black-box audit: each listener's change under the swap against its own variation across wordings (S = X - W)

Preregistered, replayable evidence for one structured handoff field

Deriving the field from public data does not stop a listener that holds the convention

Source: README, Related work

Where the evidence stands

Supported in this setting

- A planted two-ref codeword crossed a valid handoff.
- It changed the coalition listener's allocations versus public refs.
- The paired audit separated instructed listener types.

Still open

- Spontaneous convention formation.
- Unknown carrier field detection.
- Task-valid second-model transfer.
- Production incidence or risk rate.
- S can flag legitimate use; here refs carry no task information.
- Wording effect: +0.50 to +1.20 items per answer.
- Honest listener gave one answer in all six trials on 22/30 tasks.

Code-selected refs remove sender choice; they do not neutralize a listener that holds the convention.

Sources: DogWhistle README and PR #6 results

After the sprint

1 Align the comparison	Use one output schema and closer prompt wording across listener types; preregister a new within-task contrast.
2 Test model transfer	First require task-valid output in every cell, then repeat the registered audit on an independent model family.
3 Broaden the carrier	Probe additional schema fields and neutral replacements while measuring legitimate task cost.

Full protocols, raw ledgers and replay commands

github.com/Rick-Clinton-jpg/DogWhistle

Source: DogWhistle limitations; proposed future studies